MODEL X.0

PHASE 1 / PROJECT PROPOSAL

PRICE SIGNAL

A guarded early-warning view of Colombo food-price movements

Economy and livelihoods / Cost of living and everyday economics

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github.com/IT25101414/price-signal

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1. Executive case

Proposal in one sentence. Price Signal converts the Central Bank of Sri Lanka's Daily Price Reports into a cautious, three-report-day indication of whether an unusual Dambulla-to-Colombo price gap is likely to close, while refusing to show numbers when the evidence is stale, incomplete or too weak.

Why this qualifies

The project addresses a repeated, money-sensitive decision made by small Colombo food buyers. The source is public, dated and frequently refreshed; the intelligence comes from modelling the relationship between markets, not retrieving a price. The output is designed around uncertainty and safe refusal, so a polished interface cannot hide weak evidence.

Required proposal element	Where it is answered	Evidence available at Phase 1
Problem, affected group and named user	Sections 2 and 3	Three anonymous buyer sessions plus a clearly bounded decision
What exists and why it is insufficient	Section 4	Research and product comparison, with an explicit gap
Solution and methodology	Sections 5 and 7	Explainable ridge model, time split, placebo and safety guard
Data, extraction, temporality and right to use	Section 6	Manifest, parser evidence and dated CBSL reply
How success and failure will be judged	Sections 8 and 9	Pre-declared thresholds, baselines and fallback product
Delivery plan and ownership	Section 10	Dated solo-team plan, cut list and final deliverables

Phase 1 feasibility status

Research-phase work has already reduced the main feasibility risks: the data can be collected and parsed, the Dambulla signal adds held-out forecast value, uncertainty can be calibrated, and three buyers could use the concept with some explanation. These are preliminary implementation inputs, not a claim that the Phase 2 submission is complete. Phase 2 will freeze a final build, verify public access and demonstrate the same version on real data.

Repository: https://github.com/IT25101414/price-signal

2. Problem statement and significance

The problem

Small canteens, restaurants and shops in Colombo buy vegetables several times a week while prices can change sharply. A buyer can see today's supplier or market quote, but cannot readily tell whether a movement at Dambulla - the country's major wholesale hub - is likely to reach Pettah or Narahenpita over the next few report days. Buying before a fall or after a rise can directly reduce an already thin operating margin.

Who is affected

The named user is a Colombo small-food buyer responsible for frequent vegetable purchases without an analyst or procurement system. The project does not claim a population count because no reliable count was found. Significance instead rests on the frequency of the decision, perishability, limited cash flow and the immediate effect of price changes on small businesses.

The decision

Question	Answer
What decision?	Buy the full requirement today, split the purchase, or wait and re-check.
How often?	Daily or several times per week, depending on the business.
What is used now?	Supplier quotes, current market prices, other traders, recent experience, stock and expected demand.
What is missing?	A tested view of whether Dambulla adds advance information, how large the possible move is and when not to trust it.
What changes?	The buyer gains an additional evidence source, while still confirming the actual supplier price before purchasing.

Problem boundary

- The system supports timing and checking; it never commands the user to buy, wait or sell.
- It covers only foods with a usable Dambulla wholesale series and a supported Colombo price series.
- It estimates movement over report days, not calendar days, and a three-report-day horizon may cross a weekend or holiday.
- It does not claim observed business savings; the financial evaluation is a simulation with explicit assumptions.

3. User discovery

Three research-phase sessions

Three anonymous buyers participated with consent: a small restaurant buyer on 16 October 2026, and a vegetable-shop buyer and a hotel/catering buyer on 17 October 2026. They buy vegetables three times a week to almost daily and rely mainly on suppliers, market quotes, other traders, existing stock and expected demand.

Measure	Individual scores	Median
Main-card understanding	4, 3, 4	4/5
Trust as one buying input	4, 2, 4	4/5
Usefulness for work	5, 3, 4	4/5

Observed comprehension problems

Task	Without help	With help	Not completed	What this changed
Current card	2	1	0	Explain the Dambulla-to-Colombo relationship more directly.
Estimate and range	1	1	1	Describe a possible range and explain that past coverage was about eight times in ten.
No-signal state	1	2	0	Distinguish weak evidence from no price change.
Six-month history	0	2	1	Replace technical wording and explain the common start-at-100 baseline.
Weekly report	1	2	0	Explain that blank means a required daily price was unavailable.

What we learned

Two participants described a concrete use: checking before going to market or using the signal while planning later purchases, while still confirming the supplier's current quote. The third would continue relying mainly on suppliers until the signal's accuracy was clearer. This supports the problem and the role of the product as one cautious input, but it does not prove long-term adoption, changed purchases or realised savings. The exact tested commit, devices, setting and facilitator were not recorded, and revised wording still needs retesting.

4. Existing work and the unresolved gap

What research already establishes

Work	Contribution	Why it does not solve this user's problem
Spatial integration during COVID-19 [2]	Tests wholesale-market integration with Dambulla across commodities and regions.	No daily forecast, retail coverage, buyer interface or out-of-sample decision test.
Green-chilli spatial linkages [3]	Finds a central price-discovery role for Dambulla and slow adjustment to equilibrium.	One commodity, historical analysis and no operational warning.
Dambulla ARIMA/SARIMA study [4]	Forecasts monthly Dambulla prices from their own history.	Does not use Dambulla to anticipate Colombo or operate at daily cadence.

What products already provide

Product	What it provides	Remaining gap
CBSL Daily Price Report [1]	Authoritative daily wholesale and retail prices plus a short comparison.	PDF format, one-step historical comparison, no forecast, calibrated uncertainty or refusal policy.
HARTI publications [5]	Weekly and monthly food-price tables for Colombo and suburbs.	Historical reporting at a slower cadence; no source-to-destination signal.
FAO FPMA [6]	Price scanning, comparisons and anomaly views across series.	An anomaly in one series does not establish whether a source market leads a destination market.

The gap Price Signal fills

The missing product is not another price table. It is a tested answer to a relational question: after accounting for a Colombo price's own recent behaviour, does the same item's Dambulla price add useful information about the next few Colombo report days? Price Signal measures that incremental value on unseen dates, reports uncertainty, tests a placebo, and presents a refusal instead of a number when the evidence is not fit for use.

Prior ModelX entries were not available for review, so the proposal makes no claim of novelty relative to them. Novelty is claimed only relative to the literature and products reviewed above.

5. Proposed solution and user experience

Product promise

For each supported item and selected Colombo series, the product shows the latest Dambulla and Colombo prices, how unusual their gap is relative to the trailing history, the estimated Colombo change over the next three report days, an 80% possible range, a plain strength label and the data date. It says 'no clear signal' or gives a refusal reason when an estimate is not justified.

The six-page interface

Page	Purpose
Welcome	Non-technical introduction, intended user, independence and non-endorsement notice.
Today	Source health, signal counts, filters, strongest-signal navigation, uncertainty view and detailed cards.
Explore	Paired historical prices and their gap, with shared-date and start-at-100 explanations.
Weekly report	Two complete five-report-day windows, guarded CSV export and explanations for blank values.
How it works	Saved evaluation evidence, plain-language limits and traceable source links.
Help	Plain-language labels, limitations and safe-refusal explanations.

One-card reading order

1. Check the source state and report date.
2. Select the Colombo market and an item the buyer knows.
3. Read the current Dambulla and Colombo prices and the unusual-gap statement.
4. Read the middle estimate together with its possible range; neither is a guaranteed future price.
5. If the guard refuses, use the explanation and confirm the market directly.

Value without overclaiming

The user could not reproduce this insight from a single PDF or supplier call. It requires years of aligned prices, a rolling normal market gap, a time-respecting comparison with the price's own history, and uncertainty calibrated on outcomes the model did not see. The product deliberately stops short of automating the purchase decision because urgency, stock, spoilage, transport and cash flow are not in the dataset.

6. Data: source, extraction, rights and temporality

Source and inventory

Element	Verified position at 18 October 2026
Publisher	Statistics Department, Central Bank of Sri Lanka (CBSL)
Dataset	Daily Price Report PDFs: wholesale and retail prices by item and market
Manifest	2,274 discovered records through the 15 October report; 2,273 downloaded successfully and one recorded failure
Supported analysis	1,487 report days from 28 July 2020 onward; older layouts are excluded rather than guessed
Public build	Cards and a one-year history slice through 15 October 2026; raw PDFs and the full generated table are not redistributed
Personal data	None; the source contains prices by item, market and date

Right to use

CBSL's Statistics Department replied in writing on 12 October 2026 that the published data are publicly available and do not require specific approval for use. It requested attribution to the Statistics Department and stated that CBSL does not endorse or accept responsibility for derived findings. A redacted copy is retained in the public repository. The collector also follows robots.txt, uses the required ten-second delay and stops if access rules change.

Extraction and quality controls

- A resumable collector downloads each PDF once and records URL, report date, file hash, status and download time.
- The parser uses horizontal word positions because the PDF text layer drops empty cells and can shift values between markets.
- Unknown layouts, scans without text, undated headers and exact duplicates are rejected and logged.
- Repository-owned unit tests check cell parsing, row grouping, name cleaning and cross-report consistency; a separate 125-value check uses locally held original PDFs.
- The parser keeps 'n.a.', blank/not-applicable cells and market closure as distinct states; it never silently fills them.

Temporality

Historical and current data serve different purposes. Reports through 2023 train the evaluation models; 2024 onward is held out for testing. The latest accepted report drives the product. If the listing cannot be checked, a newer report remains unread, or the last good report is more than three working days old, the interface withholds estimates. A static archive would not meet a buyer's daily decision, while a short recent window would be too small to test the relationship honestly.

7. Methodology and system design

Modelling question

For wholesale Pettah, retail Pettah and retail Narahenpita, the target is the log price change over one, three or five future report days; the production product uses three. Model C uses only the Colombo series' own recent changes and distance from its 20-day mean. Model D adds Dambulla's recent changes and the Dambulla-minus-Colombo gap relative to its trailing 60-day average. The gain from C to D is the incremental information contributed by Dambulla.

Why ridge regression

The error-correction gap has a direct interpretation, the pooled dataset is large enough for a small parameter set, and ridge regularisation controls unstable coefficients without hiding the mechanism. A gradient-boosting challenger is admitted only if its three-day improvement is at least 0.5 points in every Colombo series and every confidence interval is above zero. Preliminary testing did not meet that rule, so the simpler model remains the proposal's production choice.

End-to-end architecture



Unknown layouts and stale, incomplete or weak evidence stop at the guard.

No future leakage

- Training ends in 2023 and testing begins in 2024; labels crossing the split are removed.
- Robustness intervals resample complete months rather than isolated rows.
- The placebo substitutes another item's Dambulla series and should add no skill.
- The 80% range is split-conformal: held-out residuals are scaled by recent volatility before the quantile is selected.
- Thresholds for 'no clear signal' and simulated actions are selected on training data only.

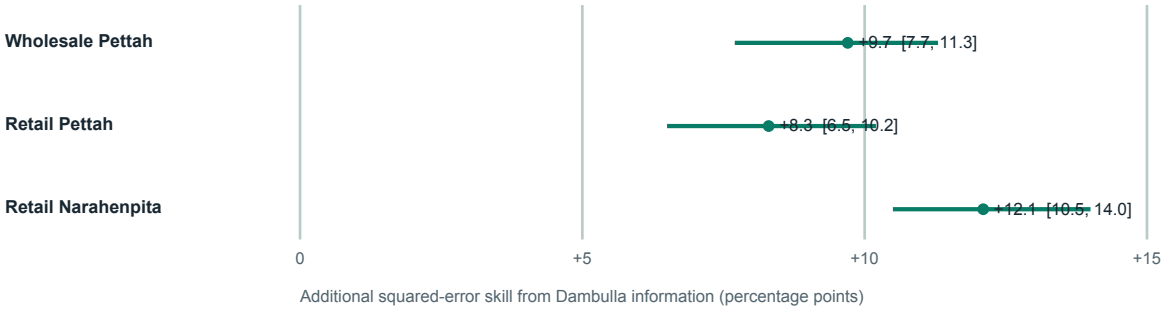
Safety guard

The guard runs when cards are created and again on every web request. It removes values for an unavailable listing, unread newer report, stale data, closed market, missing Dambulla or Colombo price, insufficient paired history or a gap too small to support a signal. This is a product requirement, not a cosmetic warning.

8. Preliminary evidence and interpretation

The following research-phase results establish feasibility and set expectations for Phase 2. They are not a substitute for the final implementation evaluation, which will be rerun on the submitted data cut.

Incremental forecast value



Latest three-day point estimates remove an additional 9.7, 8.3 and 12.1 percentage points of squared forecast error beyond the own-history model. The month-resampled robustness intervals shown above are from the matched robustness run; all remain above zero. The item-swapped placebo is approximately zero. The effect is useful but modest: root-mean-square error improves by roughly 4% to 8% rather than eliminating forecast error.

Uncertainty and simulated decision value

Colombo series	Inside stated 80% range	Average range half-width	Simulated cost vs buy-today
Wholesale Pettah	81.0%	+/- 28.1%	-1.42%
Retail Pettah	81.5%	+/- 23.3%	-1.22%
Retail Narahenpita	77.4%	+/- 18.1%	-1.57%

The nominal ranges meet the pre-declared 75% to 85% acceptance band, but they are intentionally wide. The simulated policy assumes a purchase can be delayed without shortage, spoilage, transport or operating cost. Items are reported in different units, so the project does not combine them into a misleading mixed-unit rupee total. These results are not observed buyer savings.

Negative result retained

The first idea - predicting whether a one-day jump would persist - did not beat the majority baseline (roughly 57% persistence; model accuracy 56.6% to 56.7%, AUC 0.50 to 0.54). It was rejected and remains documented. Keeping the negative result demonstrates that the project follows evidence rather than selecting only successful experiments.

9. Evaluation plan and failure conditions

What success means

Dimension	Metric and comparator	Acceptance rule
Forecast insight	Squared-error skill of Model D versus Model C, no-change and placebo; time-ordered test dates	Gain interval above zero in most series, placebo under half the real gain, and at least 60% of items improve
Uncertainty	Observed coverage of the nominal 80% range, by series and year	75% to 85%; investigate or widen outside this band
Decision relevance	Simulated purchase cost versus buy-today, always-wait, own-history and hindsight	Report effect and interval; never translate simulation into guaranteed savings
Extraction	Repository-owned parser tests plus the 125-value original-report validation	All reproducible tests pass; discrepancies are logged, never silently repaired
Safe failure	Twelve deliberately constructed faults	12/12 must refuse or pass exactly as specified
User comprehension	Task completion and 1-5 understanding, trust and usefulness ratings	Report individual scores and difficulties; revise any repeated high-severity misunderstanding

Explicit stop rules

- If the gain interval includes zero for most series, or the placebo exceeds half the real gain, stop claiming a Dambulla lead.
- If interval coverage falls below 70% or above 90%, withdraw the range until recalibrated.
- If any refusal case fails, do not publish the affected output.
- If buyers cannot explain the main card, simplify the product before using it in the video or live pitch.

Fallback product

If the predictive claim fails, the project will ship an honest descriptive version: the latest verified data date, the paired Dambulla-to-Colombo gap and a clear 'no reliable signal' state, with no forecast. This preserves useful data work without presenting an unsupported insight.

Known limits to report regardless of outcome

The study covers 18 to 19 foods, one principal train/test split and a relatively calm 2024-2026 test period. Prices are rounded; report-day horizons cross weekends and holidays; the buyer sample is three; the revised wording has not been retested; and market structure may change during a crisis. The proposal treats these as boundaries, not footnotes to hide.

10. Implementation plan, risks and deliverables

Milestones

Dates	Milestone	Done when
10-19 Oct	Research, data pipeline, parser, feasibility evidence, user discovery and proposal	Proposal submitted; repository public; claims trace to saved evidence
20-26 Oct	Freeze the final data cut, rerun modelling and calibration, close high-risk gaps	Metrics reproduce; 85+ tests and all refusal cases pass
27-29 Oct	Public deployment, wording retest if available, accessibility and signed-out checks	Every page and /healthz work without authentication
30-31 Oct	Technical report, two-page summary and three-minute video from the same build	Files agree on commit, data date, claims and limitations
1 Nov	Implementation submission	Repository, website, report, summary and video open in a private browser
9-14 Nov	If selected, rehearse the live final	Live demo runs on current real data and safe refusal can be explained

Priority risks and controls

Risk	Control / contingency
Signal weakens or was luck	Rerun time-ordered tests and placebo; use the descriptive fallback if the claim fails.
Market regime changes	Track results by year; refuse incomplete inputs; state that the calm test period may not generalise to crises.
CBSL layout or access changes	Respect current rules, stop on access changes, quarantine unknown layouts and show the last verified date.
Users treat it as advice	No buy/sell instruction, prominent uncertainty, non-endorsement, repeated server-side guard and comprehension testing.
Solo-team time pressure	Cut challenger models, translations, cloud scheduling and secondary horizons before any safety or evaluation work.

Phase 2 deliverables

- Public Git repository with a README that runs from a clean machine, pinned dependencies and visible history.
- Public web application using real CBSL data, plus safe local fallback instructions.
- Technical report evaluating the implementation against this proposal, including gaps and negative results.
- Two-page summary and a three-minute video demonstrating the exact submitted build end to end.

Ownership and cut list

Nivetha owns data, modelling, interface, evaluation, documentation and delivery. If time runs short, cut in this order: challenger models, Sinhala/Tamil versions, automated cloud refresh, one- and five-day views, then extra buyer sessions. Never cut freshness, refusal rules, data-rights evidence or honest evaluation.

11. References, disclosure and team

References

[1] Statistics Department, Central Bank of Sri Lanka. Daily Price Report. https://www.cbsl.gov.lk/en/statistics/economic-indicators/price-report. Accessed 18 Oct 2026.

[2] Spatial Integration of Vegetable Markets of Sri Lanka during COVID-19 Pandemic. Tropical Agricultural Research 33(1), 2021. https://tar.sljol.info/articles/8538

[3] Spatial linkages of green chili prices between a dedicated economic centre and other regional markets in Sri Lanka. Sri Lanka Journal of Food and Agriculture 4(1). https://sljfa.sljol.info/articles/56

[4] A comparison of price fluctuations for green chilli and tomato in Dambulla Dedicated Economic Center in Sri Lanka. Fifth Annual Science Research Sessions, South Eastern University of Sri Lanka, 2016. https://dspace.lib.seu.ac.lk

[5] Hector Kobbekaduwa Agrarian Research and Training Institute. Agricultural Commodity Prices. https://dl-harti.nsf.gov.lk

[6] FAO GIEWS. Food Price Monitoring and Analysis Tool. https://www.fao.org/giews/food-prices/price-tool

[7] Director, Statistics Department, Central Bank of Sri Lanka. Email reply to N. Ramanan, 12 Oct 2026. Redacted copy: https://github.com/IT25101414/price-signal/blob/main/docs/cbsl_reply_2026-10-12.pdf

Ethics, licensing and responsibility

- No personal data are used. Buyer records are anonymised and published only with permission.
- The original software is MIT-licensed. CBSL data, reports, the Sri Lankan emblem and other third-party material remain under their respective terms.
- Attribution is to the Statistics Department, Central Bank of Sri Lanka. CBSL does not endorse the project or accept responsibility for its derived findings.
- The project author owns every submitted claim and will not describe simulated savings as observed impact.

AI assistance disclosure

Claude (Anthropic) and OpenAI Codex assisted with research, first drafts of code and documents, testing, review, safety hardening, interface work and document production. The repository's AI_USE.md gives the file-level account. The author ran, reviewed and accepts responsibility for the system, evidence and submission.

Team details

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Project / team name	Price Signal
Repository	https://github.com/IT25101414/price-signal

Declaration: This proposal distinguishes preliminary evidence from final implementation claims, states the limits of the data and evaluation, and identifies a safe fallback if the proposed predictive insight does not hold.